PROJECT DOCUMENTATION

Railway Ticket Reservation System

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1. ABSTRACT

The Railway Ticket Reservation System is a web-based application developed using ASP.NET Web Forms and SQL Server. The main objective of this system is to provide an efficient and user-friendly platform for booking railway tickets online. The system allows users to register, log in, select travel details, choose seats, and generate tickets. It also includes an admin module to view all bookings.

The system ensures proper validation, seat selection, and ticket confirmation, making it a simple yet effective simulation of a real-time railway reservation system.

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2. OBJECTIVE

The main objectives of this project are:

- To develop an online ticket booking system
- To implement user authentication (Register/Login)
- To provide seat selection functionality
- To generate ticket confirmation
- To manage bookings through admin panel

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3. SYSTEM OVERVIEW

This system consists of two main modules:

User Module

- Register account
- Login

- Book tickets
- Select seats
- View ticket

Admin Module

- Login as admin
- View all bookings

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4. SYSTEM FLOW

Home → Register → Login → Validate → (User / Admin)
User → Booking → Confirmation → Logout
Admin → View Bookings → Logout

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5. TECHNOLOGY USED

- Frontend: HTML, CSS
- Backend: ASP.NET (C#)
- Database: SQL Server
- IDE: Visual Studio

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6. DATABASE DESIGN

Users Table

Column Name	Data Type
UserId	INT (Primary Key)
Username	VARCHAR
Password	VARCHAR

Role	VARCHAR

Tickets Table

Column Name	Data Type
TicketId	INT (Primary Key)
UserId	INT
Source	VARCHAR
Destination	VARCHAR
TravelDate	DATE
SeatNo	VARCHAR

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7. MODULE DESCRIPTION

Home Page

- Provides navigation to Register and Login
- Contains basic UI and animation

Code :

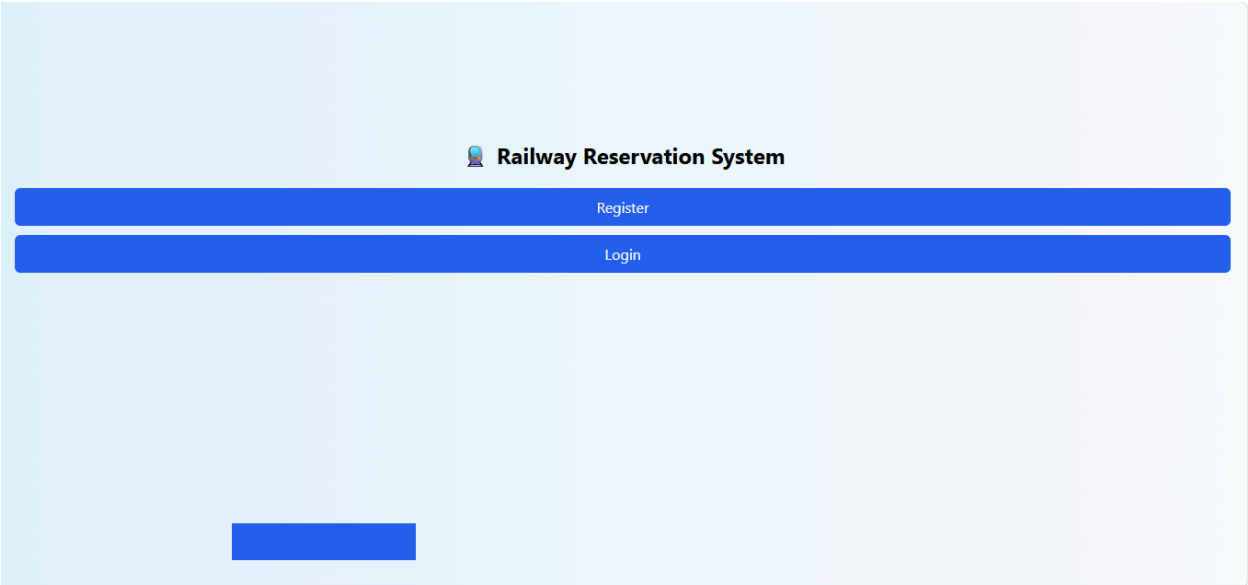
Aspx code:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Home.aspx.cs"
Inherits="TicketReservationSystem.Home" %>

<!DOCTYPE html>

<html>

<head runat="server">
```

```
<title>Home</title>


<style>
body {
    font-family:'Segoe UI';
    background:linear-gradient(to right,#e0f2fe,#f8fafc);
    text-align:center;
    overflow:hidden;
}
/* Train Animation */
.train {
    position:absolute;
    bottom:30px;
    left:-200px;
    width:200px;
    height:40px;
    background:#2563eb;
    animation:moveTrain 8s linear infinite;
}
@keyframes moveTrain {
    from { left:-200px; }
    to { left:100%; }
}
```

```
.card {
    margin-top:150px;
}
a {
    display:block;
    margin:10px;
    padding:10px;
    background:#2563eb;
    color:white;
    text-decoration:none;
    border-radius:6px;
}
</style>
</head>
<body>
<div class="train"></div>
<div class="card">
<h2>🚆 Railway Reservation System</h2>
<a href="Register.aspx">Register</a>
<a href="Login.aspx">Login</a>
</div>
</body>
</html>
```

Output :



Register Page

- Allows new users to create account
- Stores data in Users table

Code

Aspx code:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Register.aspx.cs"
Inherits="TicketReservationSystem.Register" %>

<!DOCTYPE html>

<html>

<head runat="server">

<title>Register</title>



<style>

body {
```

```
    font-family:'Segoe UI';
    background:#f1f5f9;
    display:flex;
    justify-content:center;
    align-items:center;
    height:100vh;
}
.card {
    background:white;
    padding:25px;
    border-radius:10px;
    width:300px;
    box-shadow:0 5px 15px rgba(0,0,0,0.2);
}
input, select {
    width:100%;
    padding:8px;
    margin:10px 0;
}
.btn {
    background:#2563eb;
    color:white;
    border:none;
    padding:10px;
    width:100%;
}
</style>
```

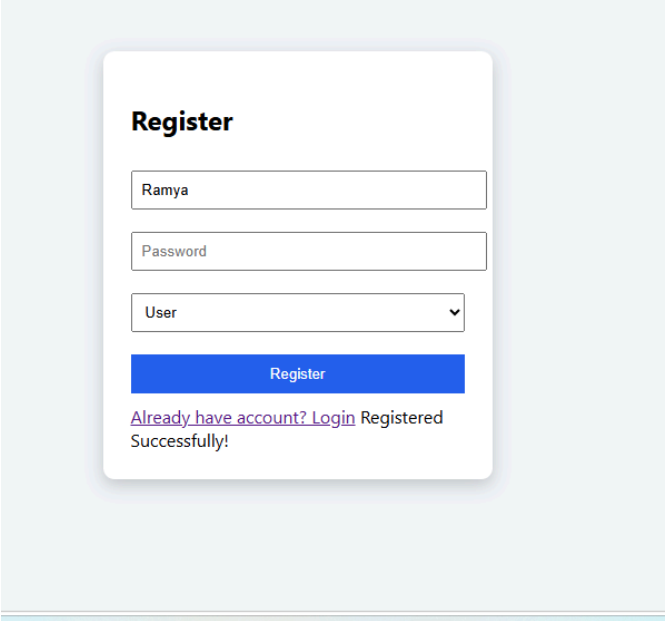
```
</head>

<body>

<form runat="server">

<div class="card">

<h2>Register</h2>

<asp:TextBox ID="txtUser" runat="server" placeholder="Username"></asp:TextBox>

<asp:TextBox ID="txtPass" runat="server" TextMode="Password" placeholder="Password"></asp:TextBox>

<asp:DropDownList ID="ddlRole" runat="server">

   <asp:ListItem Text="User" Value="User" />

   <asp:ListItem Text="Admin" Value="Admin" />

</asp:DropDownList>

<asp:Button ID="btnRegister" runat="server" Text="Register" CssClass="btn"
OnClick="btnRegister_Click"/>

   <br />

<a href="Login.aspx">Already have account? Login</a>

<asp:Label ID="lblMsg" runat="server"></asp:Label>

</div>

</form>

</body>

</html>
```

Aspx cs code :

```
using System;

using System.Data.SqlClient;

using System.Configuration;

namespace TicketReservationSystem

{

   public partial class Register : System.Web.UI.Page

   {
```

```
    protected void btnRegister_Click(object sender, EventArgs e)
    {
        SqlConnection con = new SqlConnection(
            ConfigurationManager.ConnectionStrings["dbcon"].ConnectionString);
        con.Open();
        SqlCommand cmd = new SqlCommand(
            "INSERT INTO Users (Username, Password, Role) VALUES (@u,@p,@r)", con);
        cmd.Parameters.AddWithValue("@u", txtUser.Text);
        cmd.Parameters.AddWithValue("@p", txtPass.Text);
        cmd.Parameters.AddWithValue("@r", ddlRole.SelectedValue);
        cmd.ExecuteNonQuery();
        con.Close();
        lblMsg.Text = "Registered Successfully!";
    }
  }
}
```

Output :



Login Page

- Validates user credentials
- Redirects based on role (User/Admin)

Code :

Aspx code:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Login.aspx.cs"
Inherits="TicketReservationSystem.Login" %>

<!DOCTYPE html>

<html>

<head runat="server">

<title>Login</title>

<style>

body {

   font-family:'Segoe UI';

   background:#e2e8f0;

   display:flex;

   justify-content:center;

   align-items:center;

   height:100vh;

}

.card {

   background:white;

   padding:25px;

   border-radius:10px;

   width:300px;

   box-shadow:0 5px 15px rgba(0,0,0,0.2);

}
```

```
input {
    width:100%;
    padding:8px;
    margin:10px 0;
}
.btn {
    background:#2563eb;
    color:white;
    border:none;
    padding:10px;
    width:100%;
}
</style>
</head>
<body>
<form runat="server">
<div class="card">
<h2>Login</h2>
<asp:TextBox ID="txtUser" runat="server" placeholder="Username"></asp:TextBox>

<asp:TextBox ID="txtPass" runat="server" TextMode="Password" placeholder="Password"></asp:TextBox>
<asp:Button ID="btnLogin" runat="server" Text="Login" CssClass="btn" OnClick="btnLogin_Click"/>
<asp:Label ID="lblMsg" runat="server"></asp:Label>
    <br />
<a href="Register.aspx">New user? Register</a>
</div>
</form>
```

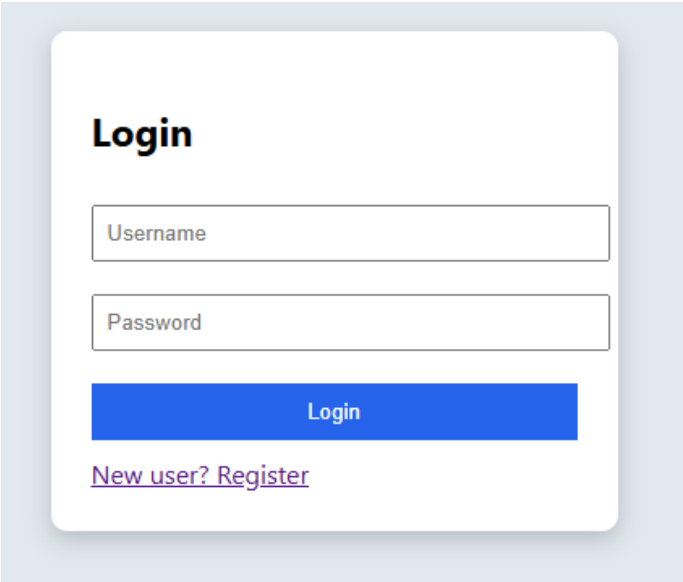
```
</body>

</html>
```

Aspx cs code :

```
using System;

using System.Data.SqlClient;

using System.Configuration;

namespace TicketReservationSystem

{

    public partial class Login : System.Web.UI.Page

    {

        protected void btnLogin_Click(object sender, EventArgs e)

        {

            SqlConnection con = new SqlConnection(

                ConfigurationManager.ConnectionStrings["dbcon"].ConnectionString);

            con.Open();


            SqlCommand cmd = new SqlCommand(

                "SELECT UserId, Role FROM Users WHERE Username=@u AND Password=@p", con);

            cmd.Parameters.AddWithValue("@u", txtUser.Text);

            cmd.Parameters.AddWithValue("@p", txtPass.Text);

            SqlDataReader dr = cmd.ExecuteReader();

            if (dr.Read())

            {

                // Store session

                Session["uid"] = dr["UserId"].ToString();

                Session["role"] = dr["Role"].ToString();
```

```
            // Role-based redirect
            if (dr["Role"].ToString() == "Admin")
                Response.Redirect("Admin.aspx");
            else
                Response.Redirect("Booking.aspx");
        }
        else
        {
            lblMsg.Text = "Invalid Login!";
        }
        con.Close();
    }
  }
}
```

Output :



Booking Page

- User selects source, destination, date
- Selects seats
- Stores booking in database

Code :

Aspx code:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Booking.aspx.cs"
Inherits="TicketReservationSystem.Booking" %>


<!DOCTYPE html>

<html>

<head runat="server">

<title>Train Booking</title>


<style>

body {

   font-family:'Segoe UI';

   background:#eef2ff;

   text-align:center;

}


/* Card */

.container {

   background:white;

   width:450px;

   margin:30px auto;

   padding:20px;

   border-radius:10px;
```

```
    box-shadow:0 5px 15px rgba(0,0,0,0.2);
}

/* Coach Layout */
.coach {
    display:grid;
    grid-template-columns: repeat(3,50px) 20px repeat(2,50px);
    gap:10px;
    justify-content:center;
    margin-top:20px;
}

.seat {
    width:50px;
    height:50px;
    background:#c7d2fe;
    display:flex;
    align-items:center;
    justify-content:center;
    border-radius:6px;
    cursor:pointer;
}

.seat:hover {
    background:#6366f1;
    color:white;
}
```

```
.seat.selected {
    background:#22c55e;
    color:white;
}

.booked {
    background:red !important;
    cursor:not-allowed;
}

.gap {
    width:20px;
}

.btn {
    padding:10px;
    background:#2563eb;
    color:white;
    border:none;
    border-radius:6px;
}
</style>

<script>
    function selectSeat(el) {
        if (el.classList.contains("booked")) return;
```

```
        el.classList.toggle("selected");

        let seats = document.querySelectorAll(".seat.selected");
        let arr = [];

        seats.forEach(s => arr.push(s.innerText));

        document.getElementById("<%= hiddenSeats.ClientID %>").value = arr.join(",");
    }
</script>

</head>

<body>

<form runat="server">

<div class="container">

<h2>🚆 Train Booking</h2>

<!-- Source -->
<asp:DropDownList ID="ddlSource" runat="server">
    <asp:ListItem>Chennai</asp:ListItem>
    <asp:ListItem>Madurai</asp:ListItem>
    <asp:ListItem>Trichy</asp:ListItem>
```

```
</asp:DropDownList>

<!-- Destination -->
<asp:DropDownList ID="ddlDest" runat="server">
    <asp:ListItem>Bangalore</asp:ListItem>
    <asp:ListItem>Coimbatore</asp:ListItem>
    <asp:ListItem>Hyderabad</asp:ListItem>
</asp:DropDownList>

<br /><br />

<!-- Date -->
<asp:Calendar ID="calDate" runat="server"></asp:Calendar>

<h3>Select Seats</h3>

<div class="coach">
    <div class="seat" id="seat1" onclick="selectSeat(this)">1</div>
    <div class="seat" id="seat2" onclick="selectSeat(this)">2</div>
    <div class="seat" id="seat3" onclick="selectSeat(this)">3</div>

    <div class="gap"></div>

    <div class="seat" id="seat4" onclick="selectSeat(this)">4</div>
    <div class="seat" id="seat5" onclick="selectSeat(this)">5</div>

    <div class="seat" id="seat6" onclick="selectSeat(this)">6</div>
```

```
    <div class="seat" id="seat7" onclick="selectSeat(this)">7</div>

    <div class="seat" id="seat8" onclick="selectSeat(this)">8</div>


    <div class="gap"></div>


    <div class="seat" id="seat9" onclick="selectSeat(this)">9</div>

    <div class="seat" id="seat10" onclick="selectSeat(this)">10</div>

</div>


<asp:HiddenField ID="hiddenSeats" runat="server" />


<br />


<asp:Button ID="btnBook" runat="server" Text="Book Ticket" CssClass="btn" OnClick="btnBook_Click"/>


<br />


<asp:Label ID="lblMsg" runat="server"></asp:Label>


</div>


</form>


</body>

</html>
```

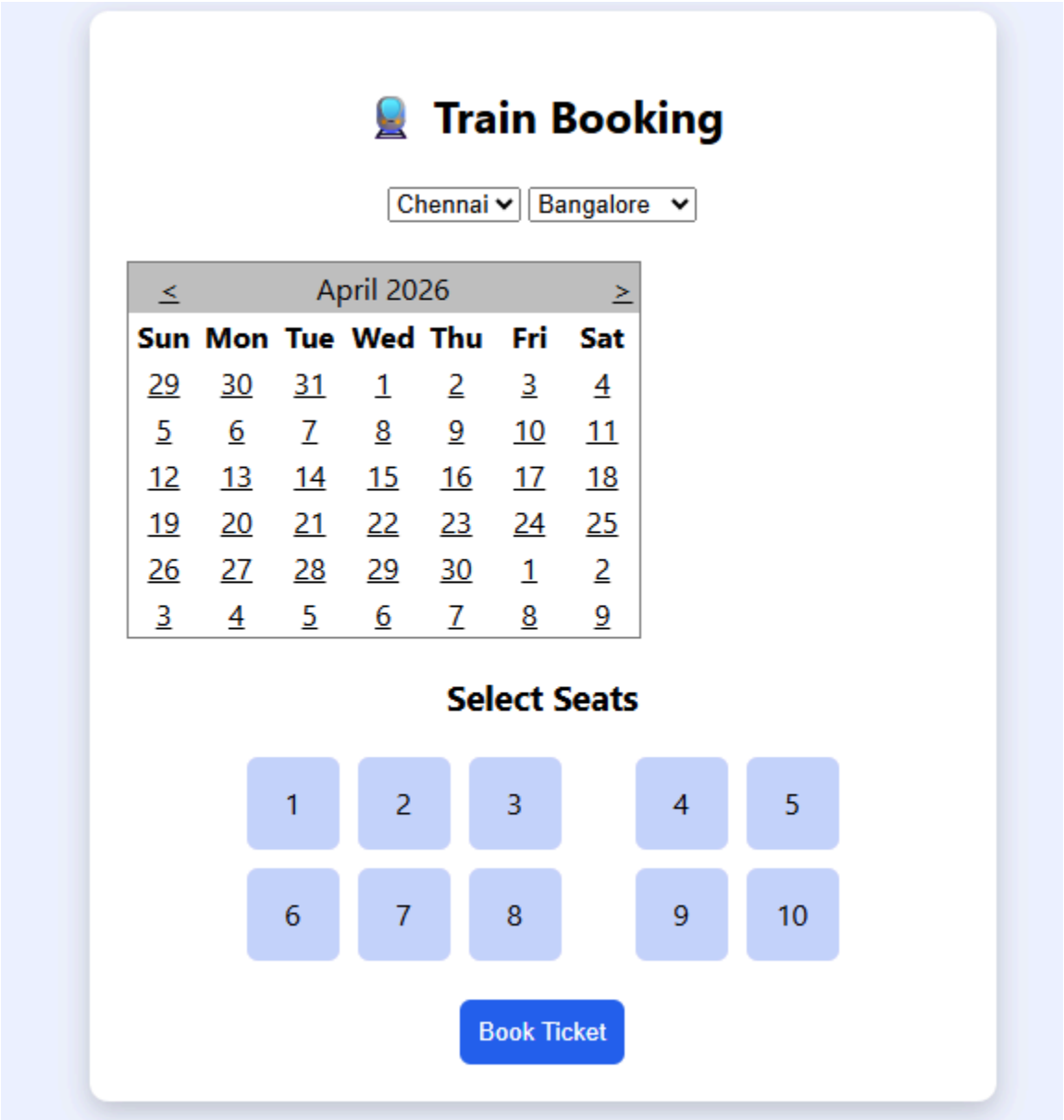
Aspx cs code :

```
using System;

using System.Data.SqlClient;

using System.Configuration;


namespace TicketReservationSystem

{

    public partial class Booking : System.Web.UI.Page

    {

        protected void btnBook_Click(object sender, EventArgs e)

        {

            if (ddlSource.SelectedValue == ddlDest.SelectedValue)

            {

                lblMsg.Text = "Source and Destination cannot be same!";

                return;

            }


            if (hiddenSeats.Value == "")

            {

                lblMsg.Text = "Select seats!";

                return;

            }


            SqlConnection con = new SqlConnection(

                ConfigurationManager.ConnectionStrings["dbcon"].ConnectionString);


            con.Open();
```

```
        string[] seats = hiddenSeats.Value.Split(',');

        foreach (string seat in seats)
        {
            SqlCommand cmd = new SqlCommand(
                "INSERT INTO Tickets (UserId, Source, Destination, TravelDate, SeatNo) VALUES
(@u,@s,@d,@date,@seat)", con);

            cmd.Parameters.AddWithValue("@u", Session["uid"]);
            cmd.Parameters.AddWithValue("@s", ddlSource.SelectedValue);
            cmd.Parameters.AddWithValue("@d", ddlDest.SelectedValue);
            cmd.Parameters.AddWithValue("@date", calDate.SelectedDate);
            cmd.Parameters.AddWithValue("@seat", seat);

            cmd.ExecuteNonQuery();
        }

        con.Close();

        // ✅ IMPORTANT PART
        Session["source"] = ddlSource.SelectedValue;
        Session["dest"] = ddlDest.SelectedValue;
        Session["seats"] = hiddenSeats.Value;
        Session["date"] = calDate.SelectedDate.ToShortDateString();

        Response.Redirect("Confirmation.aspx");
    }
```

```
  }
}
```

Output :


Train Booking
Chennai
Bangalore
<
April 2026
>
Sun Mon Tue Wed Thu Fri Sat
29 30 31 1 2 3 4
5 6 7 8 9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 1 2
3 4 5 6 7 8 9
Select Seats
1 2 3 4 5
6 7 8 9 10
Book Ticket

Confirmation Page

- Displays ticket details
- Allows printing of ticket

Code :

Aspx code:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Confirmation.aspx.cs"
Inherits="TicketReservationSystem.Confirmation" %>


<!DOCTYPE html>

<html>

<head runat="server">

<title>Ticket</title>


<style>

body {

   font-family:'Segoe UI';

   background:#e2e8f0;

   display:flex;

   justify-content:center;

   align-items:center;

   height:100vh;

}


/* Ticket Card */

.ticket {

   width:400px;

   background:white;
```

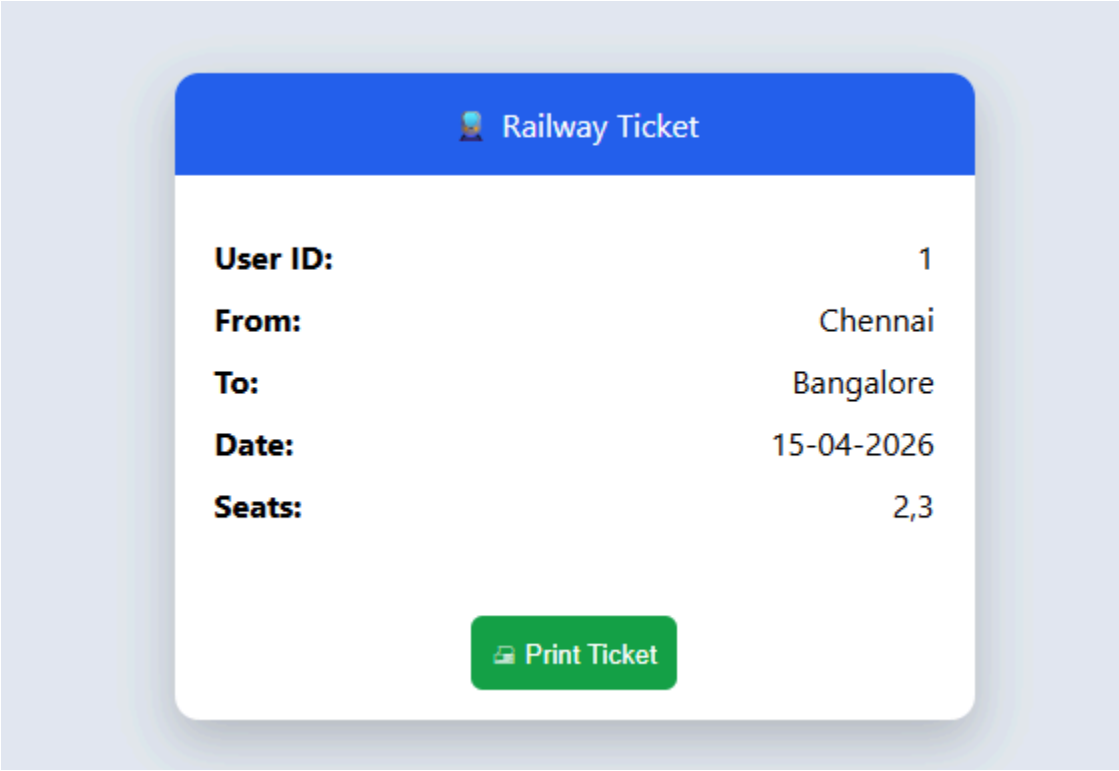
```
    border-radius:12px;
    box-shadow:0 10px 25px rgba(0,0,0,0.2);
    overflow:hidden;
}

/* Header */
.header {
    background:#2563eb;
    color:white;
    padding:15px;
    text-align:center;
}

/* Content */
.content {
    padding:20px;
}

.row {
    display:flex;
    justify-content:space-between;
    margin:10px 0;
}

/* Footer */
.footer {
    text-align:center;
```

```
    padding:15px;
}

.btn {
    padding:10px;
    background:#16a34a;
    color:white;
    border:none;
    border-radius:6px;
}
</style>
</head>

<body>

<form runat="server">

<div class="ticket">

    <div class="header">
        🚆 Railway Ticket
    </div>

    <div class="content">

        <div class="row">
            <b>User ID:</b>
```

```
        <span><%= Session["uid"] %></span>
    </div>

    <div class="row">
        <b>From:</b>
        <span><%= Session["source"] %></span>
    </div>

    <div class="row">
        <b>To:</b>
        <span><%= Session["dest"] %></span>
    </div>

    <div class="row">
        <b>Date:</b>
        <span><%= Session["date"] %></span>
    </div>

    <div class="row">
        <b>Seats:</b>
        <span><%= Session["seats"] %></span>
    </div>

</div>

<div class="footer">
    <button class="btn" onclick="window.print()">🖨️ Print Ticket</button>
```

```
    </div>

</div>

</form>

</body>

</html>
```

Output :



Admin Page

- Displays all bookings
- Uses GridView for data display

Code :

Aspx code:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Admin.aspx.cs"
Inherits="TicketReservationSystem.Admin" %>

<!DOCTYPE html>

<html>

<head runat="server">

<title>Admin</title>

<style>

body {

   font-family:'Segoe UI';

   background:#f1f5f9;

   text-align:center;

}

.grid {

   margin:30px auto;

   width:80%;

}

</style>

</head>

<body>

<form runat="server">

<h2>Admin Dashboard</h2>
```

```
<asp:GridView ID="GridView1" runat="server" CssClass="grid" />

</form>

</body>

</html>
```

Aspx cs code :

```
using System;

using System.Data;

using System.Data.SqlClient;

using System.Configuration;

namespace TicketReservationSystem

{

    public partial class Admin : System.Web.UI.Page

    {

        protected void Page_Load(object sender, EventArgs e)

        {

            // Only admin allowed

            if (Session["role"] == null || Session["role"].ToString() != "Admin")

            {

                Response.Redirect("Login.aspx");

            }

            if (!IsPostBack)

            {

                LoadData();

            }

        }

        void LoadData()

        {
```

```
        SqlConnection con = new SqlConnection(
            ConfigurationManager.ConnectionStrings["dbcon"].ConnectionString);
        SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM Tickets", con);
        DataTable dt = new DataTable();
        da.Fill(dt);


        GridView1.DataSource = dt;
        GridView1.DataBind();
    }
  }
}
```

Output :

Admin Dashboard

TicketId	UserId	Source	Destination	TravelDate	SeatNo
1	1	Chennai	Coimbatore	16-04-2026 00:00:00	1,2,3
2	1	Chennai	Coimbatore	16-04-2026 00:00:00	1,2,3
3	1	Chennai	Bangalore	08-04-2026 00:00:00	1
4	1	Chennai	Bangalore	08-04-2026 00:00:00	2
5	1	Chennai	Bangalore	08-04-2026 00:00:00	3
6	1	Chennai	Bangalore	02-04-2026 00:00:00	1
7	1	Chennai	Bangalore	02-04-2026 00:00:00	2
8	1	Chennai	Bangalore	02-04-2026 00:00:00	3
9	1	Chennai	Bangalore	02-04-2026 00:00:00	6
10	1	Chennai	Bangalore	02-04-2026 00:00:00	8
11	1	Chennai	Bangalore	02-04-2026 00:00:00	1
12	1	Chennai	Bangalore	02-04-2026 00:00:00	2
13	1	Trichy	Bangalore	07-04-2026 00:00:00	1
14	1	Trichy	Bangalore	07-04-2026 00:00:00	2
15	1	Chennai	Bangalore	30-03-2026 00:00:00	2
16	1	Chennai	Bangalore	30-03-2026 00:00:00	3
17	1	Chennai	Bangalore	30-03-2026 00:00:00	2
18	1	Chennai	Bangalore	30-03-2026 00:00:00	3
19	1	Chennai	Bangalore	30-03-2026 00:00:00	2
20	1	Chennai	Bangalore	30-03-2026 00:00:00	3
21	1	Trichy	Bangalore	02-04-2026 00:00:00	2

8. KEY FEATURES

- User Registration and Login
- Role-based access (User/Admin)
- Seat selection system
- Ticket generation
- Print ticket option
- Admin dashboard

---

9. VALIDATION

- Prevent empty inputs
- Prevent same source & destination
- Ensure seat selection
- Session-based authentication

---

10. FUTURE ENHANCEMENT

- Online payment integration
- Train schedule system
- Seat availability per date
- Email/SMS confirmation
- Advanced UI design

---

11. CONCLUSION

The Railway Ticket Reservation System successfully demonstrates a basic online booking system using ASP.NET and SQL Server. It provides essential functionalities such as registration, login, booking, and ticket generation. The project enhances understanding of web development, database connectivity, and user interface design.